

Task 1

Workflow is the automation of a process in which documents and information are passed between participants in accordance with a set of rules. (Hollingsworth, 1995) A workflow management system (WfMS) is software that defines workflows and creates and manages their execution.

A running workflow instance is one particular execution of a workflow. It has its own identity, data, and history, and many instances can run at the same time from the same workflow definition. A workflow definition is a reusable blueprint, with steps, orders, rules, and transitions from which a running workflow instance is built. Activities and workflow items are represented as runtime objects or records inside a workflow instance. (IBM, 2024)

Work items are concrete executions of tasks that are assigned to specific people, groups, or services and routed between workflow steps by the engine according to rules defined in the workflow. Approvals, rejections, and escalations are handled as explicit decision points with timeouts and alternate paths. The workflow pauses at an approval step, waits for a decision or a timeout, then follows the branch that matches the outcome. A rejection prevents continuation and usually triggers an alternate path, and then an escalation handles situations where an approval or work item is not acted on in time. (Temporal, 2026)

Participants are entities that can perform work in a workflow, such as users, groups, or organisational units, and roles are descriptions of the capabilities or responsibilities required to perform certain activities. A user can have multiple roles, and a role can be performed by multiple users. Responsibilities are the tasks or activities to be done by the users. (Hollingsworth, 1994)

Dependencies are modelled as explicit prerequisites that must be satisfied before an activity can run. Events are structured triggers that can start, advance or terminate workflow instances and failures are handled through configurable policies. (Aalst, 1998) Workflow systems' interactions with external systems are modelled as service tasks or external activities that can invoke connectors or adapters implementing a common interface. (Ferreira, 2004)

Workflow systems record history and audit information by logging structured events for every important action and state change in a workflow instance and then storing these events in a log or history tables. (List B., 2001) Running workflows are monitored and reported on by collecting execution data from the workflow engine, exposing it through dashboards, reports and alerts, and using that information to track progress, detect bottlenecks and measure performance against targets. (Dumas, 2018)

A workflow engine is reusable across different processes when it provides a generic execution core and exposes process-specific logic through external workflow definitions so that new processes can be added or modified without changing the engine itself. (Mönckemeyer, 2024)

A workflow definition includes identity and metadata, structure of the process, event and timer definitions, variable and data schema, roles and participating mappings, assignment and routing policies, error handling and escalation configuration. Whereas a running instance contains

identity and linkage, current execution state, workflow variables, work items, history and audit trail, timers and pending events, and error and exception state, which are linked to but independent of its definition. (Oracle, 2026)

In real workflow systems, nested activities or sub-processes are modelled as activities that encapsulate an internal workflow, either embedded within the parent definition or defined separately and invoked via a call activity. The engine itself stays generic, while process-specific decisions are externalised into workflow definitions, rules and configuration. The engine provides the mechanisms to evaluate and enforce those decisions, but not the decisions themselves. (OMG, 2010)

In workflow systems, changes to workflow definitions are handled by versioning the definition and binding each running instance to the version it started with, so that edits create a new version used only by new instances, while in-flight instances continue on their original version. (Lotz, 2026)

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